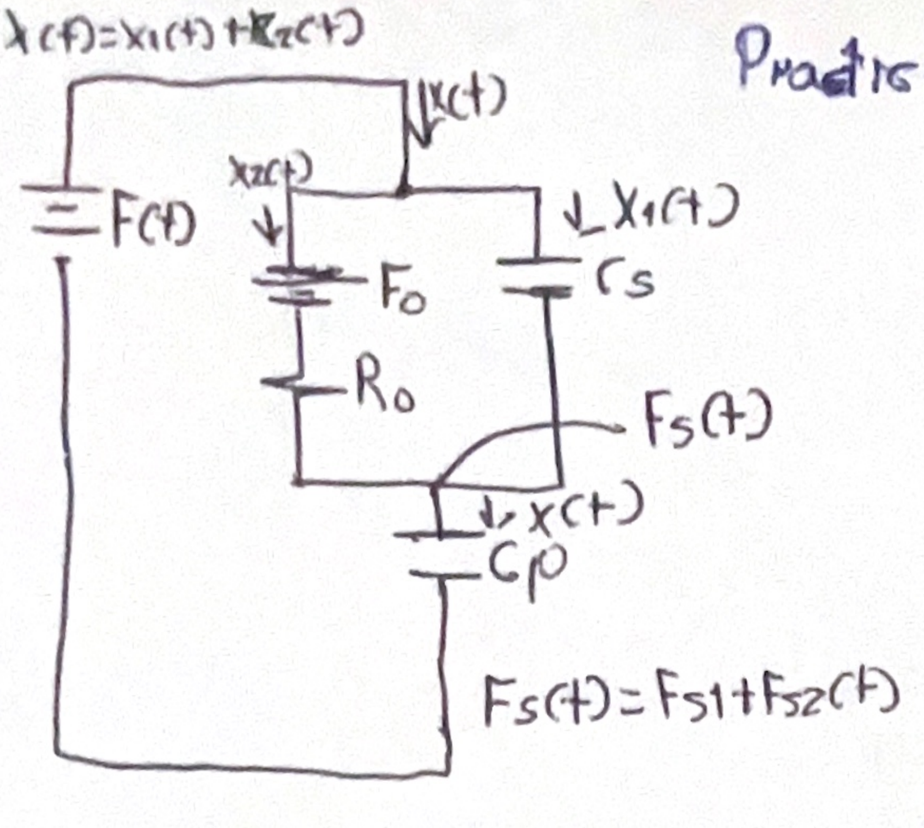
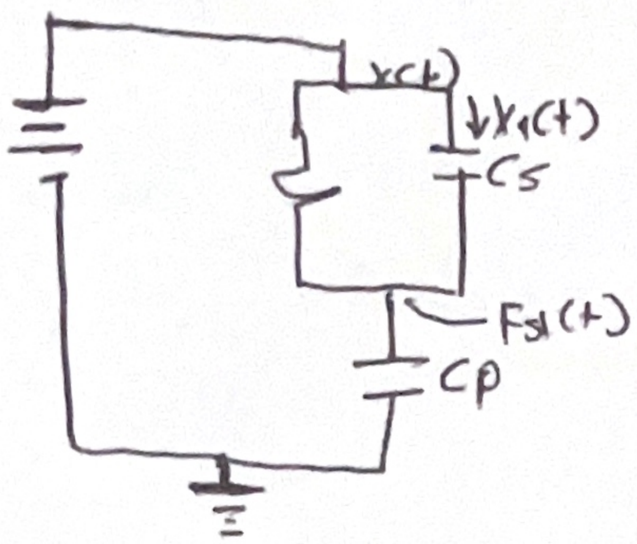
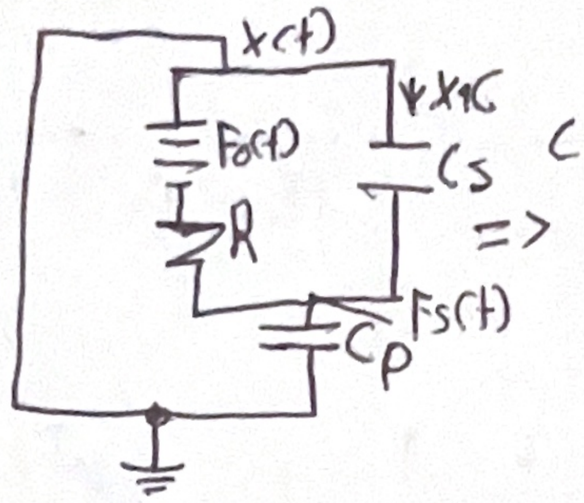
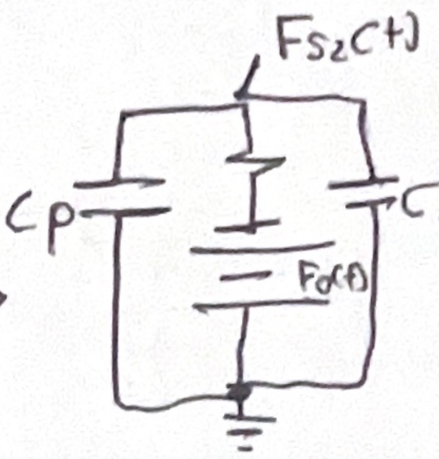
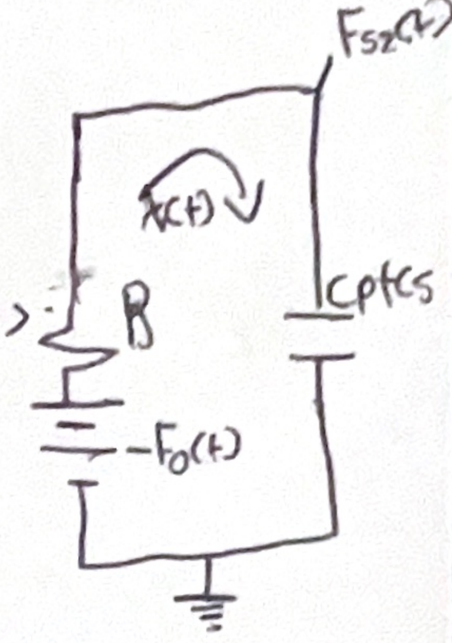
Practica 3 | Sistema musculoesquetico











Analis matematico

Debido a que el sistema contiene dos fuentes de voltaje, se debe aplicar el principio de superposicion

Circuito 1. Apagando $F_o(t)$

$$x(t)=x_1(t)+x_2(t)$$

$$x(t)=C_p\frac{dF_{s1}}{dt},\quad x_1(t)=C_s\frac{d[F(t)-F_{s1}(t)]}{dt},\quad x_2(t)=\frac{F(t)-F_{s1}(t)}{R}$$

$$C_p\frac{dF_{s1}(t)}{dt}=C_s\frac{d[F(t)-F_{s1}(t)]}{dt}+\frac{F(t)-F_{s1}(t)}{R}$$

$$C_{ps}F_{s1}(s)=C_{ss}[F(s)-F_{s1}(s)]+\frac{F(s)}{R}-\frac{F_{s1}(s)}{R}$$

$$C_{ps}F_{s1}(s)=C_sSF(s)-C_sSF_{s1}(s)+\frac{F(s)}{R}-\frac{F_{s1}(s)}{R}$$

$$C_{ps}F_{s1}(s)+C_sSF_{s1}(s)+\frac{F_{s1}}{R}=C_sSF(s)+\frac{F(s)}{R}$$

$$\left(C_{ps}+C_sS+\frac{1}{R}\right)F_{s1}(s)=\left(C_sS+\frac{1}{R}\right)F(s)$$

$$\frac{R(C_p+C_s)s+1}{R}F_{s1}(s)=\frac{C_sR_s+1}{R}F(s)$$

$$F_{s1}(s)=\frac{\frac{C_sR_s+1}{R}}{\frac{R(C_p+C_s)s+1}{R}}F(s)$$

$$F_{s1}(s)=\frac{C_sR_s+1}{R(C_p+C_s)s+1}F(s)$$

Circuito 2: Apagando F(t)

Recordando que $F_o(t) = \alpha F(t)$

$$-\alpha F(t) = R_1(t) + \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$F_{se}(t) = \frac{1}{C_p + C_s}\int x(t)\,dt$$

$$-\alpha F(s) = R x(s) + \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s2}(s) = \frac{x(s)}{(C_p + C_s)s}$$

$$F_{s1}(s) = -\frac{1}{a}\left(R + \frac{1}{(C_p + C_s)s}\right)x(s) = \frac{1}{a}\,\frac{R(C_p + C_s)s + 1}{(C_s + C_p)s}\,x(s)$$

$$\frac{F_{s2}(s)}{F(t)} = \frac{\frac{x(s)}{(C_p + C_s)s}}{-\frac{1}{a}\,\frac{R(C_p + C_s)s + 1}{(C_p + C_s)s}\,x(s)} = \frac{a}{R(C_p + C_s)s + 1}$$

Funcion de transferencia

$$F_o(s) = F_{s1}(s) + F_{s2}(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}F(s) - \frac{\alpha F(s)}{R(C_p + C_s)s + 1}$$

$$F_s(s) = \frac{C_s R s + 1}{R(C_p + C_s)s + 1}F(s)$$

$$\frac{F_s(s)}{F(s)} = \frac{C_s R s + 1 - \alpha}{R(C_p + C_s)s + 1}$$

Error en estado estacionario

$$e(s) = \lim_{s \to 0} sF(s)\left(1 - \frac{F_s(s)}{F(s)}\right) = s\frac{1}{s}\left(1 - \frac{RC_s s + 1 - \alpha}{R(C_p + RC_s)s + 1}\right) = \alpha$$

Estabilidad en lazo abierto

$$R(C_p + RC_s)s + 1 = 0$$

$$\lambda = -\frac{1}{RC_p + RC_s}$$, El sistema es estable